In[1]:=
```
realfora  = a12 ∈ Reals && a13 ∈ Reals  && a23 ∈ Reals  && a14 ∈ Reals  && a24 ∈ Reals &&
    a34 ∈ Reals  && a15 ∈ Reals  && a25 ∈ Reals  && a35 ∈ Reals  && a45 ∈ Reals  &&
    a16 ∈ Reals  && a26 ∈ Reals  && a36 ∈ Reals  && a46 ∈ Reals  && a56 ∈ Reals  &&
    a17 ∈ Reals  && a27 ∈ Reals  && a37 ∈ Reals  && a47 ∈ Reals  && a57 ∈ Reals  && a67 ∈ Reals ;
```

In[2]:=
```
sx = {{0, 1}, {1, 0}};
```

In[3]:=
```
sy = {{0, -ⅈ}, {ⅈ, 0}};
```

In[4]:=
```
sz = {{1, 0}, {0, -1}};
```

In[5]:=
```
I2 = {{1, 0}, {0, 1}};
```

In[6]:=
```
g1 = sx; g2 = sy; g3 = sz;
```

In[7]:=
```
{MatrixForm [g1], MatrixForm [g2], MatrixForm [g3]}
```

Out[7]=
$$\left\{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right\}$$

In[8]:=
```
R2 = a12 (- ⅈ g1 . g2);
```

In[9]:=
```
MatrixForm [R2]
```

Out[9]//MatrixForm=
$$\begin{pmatrix} a12 & 0 \\ 0 & -a12 \end{pmatrix}$$

In[10]:=
```
R2dual  = - (ⅈ g2) . Transpose [R2] . (ⅈ g2);
```

In[11]:=
```
MatrixForm [R2dual  + R2]
```

Out[11]//MatrixForm=
$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

In[12]:=
```
R3 = a12 (- ⅈ g1  . g2) + a13 (- ⅈ g1 . g3) + a23 (- ⅈ g2 . g3);
```

In[13]:=
```
MatrixForm [R3]
```

Out[13]//MatrixForm=
$$\begin{pmatrix} a12 & i\,a13 + a23 \\ -i\,a13 + a23 & -a12 \end{pmatrix}$$

In[14]:=
```
R3dual  = - (ⅈ g2) . Transpose [R3] . (ⅈ g2);
```

In[15]:=
```
MatrixForm [R3dual  + R3]
```

Out[15]//MatrixForm=
$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

In[16]:=
```
Eigenvalues [R3]
```

Out[16]=
$$\left\{-\sqrt{a12^2 + a13^2 + a23^2},\ \sqrt{a12^2 + a13^2 + a23^2}\right\}$$

In[17]:=
```
g1 = KroneckerProduct [sx, g1]; g2 = KroneckerProduct [sx, g2]; g3 = KroneckerProduct [sx, g3];
```

In[18]:=
```
g4 = KroneckerProduct [sy, I2];
```

In[19]:=
```
{MatrixForm [g1], MatrixForm [g2],  MatrixForm [g3], MatrixForm [g4]}
```

Out[19]=
$$\left\{\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \\ i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}\right\}$$

In[20]:=
```
z = KroneckerProduct [I2, ⅈ sy ];
```

In[21]:=
```
R4 = a12 (-i g1 . g2) + a13 (-i g1 . g3) +
    a23 (-i g2 . g3) + a14 (-i g1 . g4) + a24 (-i g2 . g4) + a34 (-i g3 . g4);
```

In[22]:=
```
MatrixForm [R4]
```

Out[22]//MatrixForm=

$$\begin{pmatrix} a12 + a34 & i\,a13 + a14 + a23 - i\,a24 & 0 & 0 \\ -i\,a13 + a14 + a23 + i\,a24 & -a12 - a34 & 0 & 0 \\ 0 & 0 & a12 - a34 & i\,a13 - a14 + a23 + i\,a24 \\ 0 & 0 & -i\,a13 - a14 + a23 - i\,a24 & -a12 + a34 \end{pmatrix}$$

In[23]:=
```
EigR4 = Simplify [ Eigenvalues [R4], Assumptions → realfora ]
```

Out[23]=

$$\left\{-\sqrt{a12^2 + a13^2 + a14^2 - 2\,a14\,a23 + a23^2 + 2\,a13\,a24 + a24^2 - 2\,a12\,a34 + a34^2},\right.$$
$$\sqrt{a12^2 + a13^2 + a14^2 - 2\,a14\,a23 + a23^2 + 2\,a13\,a24 + a24^2 - 2\,a12\,a34 + a34^2},$$
$$-\sqrt{a12^2 + a13^2 + a14^2 + 2\,a14\,a23 + a23^2 - 2\,a13\,a24 + a24^2 + 2\,a12\,a34 + a34^2},$$
$$\left.\sqrt{a12^2 + a13^2 + a14^2 + 2\,a14\,a23 + a23^2 - 2\,a13\,a24 + a24^2 + 2\,a12\,a34 + a34^2}\right\}$$

In[24]:=
```
g1 = KroneckerProduct [sx, g1]; g2 = KroneckerProduct [sx, g2];
g3 = KroneckerProduct [sx, g3]; g4 = KroneckerProduct [sx, g4];
```

In[25]:=
```
g5 = KroneckerProduct [sy, IdentityMatrix [4]];
```

In[26]:=
```
{MatrixForm [g1], MatrixForm [g2], MatrixForm [g3], MatrixForm [g4], MatrixForm [g5]}
```

Out[26]=

$$\left\{\begin{pmatrix} 0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&1&0&0 \\ 0&0&0&0&1&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0 \\ 1&0&0&0&0&0&0&0 \end{pmatrix}, \begin{pmatrix} 0&0&0&0&0&0&0&-i \\ 0&0&0&0&0&0&i&0 \\ 0&0&0&0&0&-i&0&0 \\ 0&0&0&0&i&0&0&0 \\ 0&0&0&-i&0&0&0&0 \\ 0&0&i&0&0&0&0&0 \\ 0&-i&0&0&0&0&0&0 \\ i&0&0&0&0&0&0&0 \end{pmatrix},\right.$$

$$\left.\begin{pmatrix} 0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0&-1 \\ 0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&-1&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&-1&0&0&0&0 \\ 1&0&0&0&0&0&0&0 \\ 0&-1&0&0&0&0&0&0 \end{pmatrix}, \begin{pmatrix} 0&0&0&0&0&0&-i&0 \\ 0&0&0&0&0&0&0&-i \\ 0&0&0&0&i&0&0&0 \\ 0&0&0&0&0&i&0&0 \\ 0&0&-i&0&0&0&0&0 \\ 0&0&0&-i&0&0&0&0 \\ i&0&0&0&0&0&0&0 \\ 0&i&0&0&0&0&0&0 \end{pmatrix}, \begin{pmatrix} 0&0&0&0&-i&0&0&0 \\ 0&0&0&0&0&-i&0&0 \\ 0&0&0&0&0&0&-i&0 \\ 0&0&0&0&0&0&0&-i \\ i&0&0&0&0&0&0&0 \\ 0&i&0&0&0&0&0&0 \\ 0&0&i&0&0&0&0&0 \\ 0&0&0&i&0&0&0&0 \end{pmatrix}\right\}$$

In[27]:=
```
R5 = a12 (-i g1 . g2) + a13 (-i g1 . g3) + a23 (-i g2 . g3) + a14 (-i g1 . g4) + a24 (-i g2 . g4) +
    a34 (-i g3 . g4) + a15 (-i g1 . g5) + a25 (-i g2 . g5) + a35 (-i g3 . g5) + a45 (-i g4 . g5);
```

In[28]:=
```
MatrixForm [R5]
```

Out[28]//MatrixForm=

$$\begin{pmatrix} a12 + a34 & i\,a13 + a14 + a23 - i\,a24 & a35 - i\,a45 & a15 - i\,a25 \\ -i\,a13 + a14 + a23 + i\,a24 & -a12 - a34 & a15 + i\,a25 & -a35 - i\,a45 \\ a35 + i\,a45 & a15 - i\,a25 & a12 - a34 & i\,a13 - a14 + a23 + i\,a24 \\ a15 + i\,a25 & -a35 + i\,a45 & -i\,a13 - a14 + a23 - i\,a24 & -a12 + a34 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

In[29]:=
```
EigR5 = Simplify [ Eigenvalues [R5], Assumptions → realfora ]
```

Out[29]=

$$\Big\{-\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2-2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$-\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2-2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2-2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2-2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$-\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2+2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$-\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2+2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$\sqrt{\Big(a12^2+a13^2+a14^2+a15^2+a23^2+a24^2+a25^2+a34^2+a35^2+a45^2+2\sqrt{(a13^2\,a24^2+a13^2\,a25^2-2\,a12\,a13\,a24\,a34+a12^2\,a34^2+a25^2\,a34^2+a15^2\,(a23^2+a24^2+a34^2)-2\,a12\,a13\,a25\,a35-2\,a24\,a25\,a34\,a35+a12^2\,a35^2+a24^2\,a35^2+a14^2\,(a23^2+a25^2+a35^2)+2\,a23\,a25\,a34\,a45-2\,a23\,a24\,a35\,a45+a12^2\,a45^2+a13^2\,a45^2+a23^2\,a45^2+2\,a15\,(-a13\,a23\,a25+a12\,a23\,a35+a12\,a24\,a45+a13\,a34\,a45)-2\,a14\,(a13\,a23\,a24+a15\,a24\,a25-a12\,a23\,a34+a15\,a34\,a35+a12\,a25\,a45+a13\,a35\,a45))}\Big)},$$

$$\sqrt{\left(a12^2 + a13^2 + a14^2 + a15^2 + a23^2 + a24^2 + a25^2 + a34^2 + a35^2 + a45^2 + 2\sqrt{(a13^2\, a24^2 + a13^2\, a25^2 - 2\, a12\, a13\, a24\, a34 + a12^2\, a34^2 + a25^2\, a34^2 + a15^2\,(a23^2 + a24^2 + a34^2) - 2\, a12\, a13\, a25\, a35 - 2\, a24\, a25\, a34\, a35 + a12^2\, a35^2 + a24^2\, a35^2 + a14^2\,(a23^2 + a25^2 + a35^2) + 2\, a23\, a25\, a34\, a45 - 2\, a23\, a24\, a35\, a45 + a12^2\, a45^2 + a13^2\, a45^2 + a23^2\, a45^2 + 2\, a15\,(-a13\, a23\, a25 + a12\, a23\, a35 + a12\, a24\, a45 + a13\, a34\, a45) - 2\, a14\,(a13\, a23\, a24 + a15\, a24\, a25 - a12\, a23\, a34 + a15\, a34\, a35 + a12\, a25\, a45 + a13\, a35\, a45))}\right)}$$